Schubert Calculus

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1 Grassmanians

Note:
Subvarieties are assumed to be closed.

The focus of this discussion is on Schubert calculus, in particular the sort of problems it solves in enumerative geometry. In general the problem is to determine the number of spaces (of a given dimension) intersect some set of given spaces. We are dealing here with linear spaces and thus it is natural to consider a form of topology on linear subspaces.

Definition 1.1
Let V be an n-dimensional vector space, then for $k \leq n$ we define V's **Grassmanian** to be the space

$$G(k,V) = \{\Lambda \leq V \mid \Lambda \text{ is } k\text{-dimensional}\}.$$

The Grassmanian will of course be given the structure of a variety, in due time. First of all, if $\Psi: V \cong V'$ is an isomorphism of vector spaces then their Grassmanians are in bijection

$$G(k,V) \xrightarrow{\sim} G(k,V'), \qquad \Lambda \mapsto \Psi\Lambda,$$

and this is an isomorphism of varieties. Writing

$$G(k,n) = G(k,\mathbb{F}^n),$$

we see that for n-dimensional V, $G(k,V) \cong G(k,n)$.

Note further that there is an isomorphism

$$G(k,V) \xrightarrow{\sim} G(n-k,V^*), \qquad \Lambda \mapsto \operatorname{ann}\Lambda = \{\varphi \in V^* \mid \varphi\Lambda = 0\},$$

which is again an isomorphism of varieties. Thus we have $G(k,n) \cong G(n-k,n)$.

Now we proceed with endowing $G(k,V)$ a variety structure. For $\Lambda \in G(k,V)$ consider the exterior power

$$\bigwedge^k \Lambda \leq \bigwedge^k V,$$

which is 1-dimensional. Thus there is a natural map

$$G(k,V) \to \mathbb{P}\left(\bigwedge^k V\right) \cong \mathbb{P}^{\binom{n}{k}-1}, \qquad \Lambda \mapsto \bigwedge^k \Lambda.$$

This map is called the **Plücker embedding**. We will continue to use $N = \binom{n}{k} - 1$.

Lemma 1.2
The Plücker embedding is indeed injective, and its image in $\mathbb{P}(V^k)$ is a subvariety.

Proof

Choose basis vectors $v_1, \dots, v_k \in \Lambda$ so that $\eta = v_1 \wedge \cdots \wedge v_k$ spans Λ^k. Considering the exterior algebra $\bigwedge V$, a vector $v \in V$ annihiltes η (meaning $v \wedge \eta$) iff v is in the span of $v_1, \dots, v_k$, i.e. $v \in \Lambda$. Thus we can determine Λ from its image in $\mathbb{P}(V^k)$ by

$$\Lambda = \{v \in V \mid v \wedge \eta = 0\},$$

so the Plücker embedding is injective.

Extend a basis $v_1, \dots, v_k \in \Lambda$ to a basis of V. Then we can write a $\eta \in \bigwedge^k V$ as $\eta = \sum_I \eta_I v_I$ for scalars η_I. Then

Now consider the map

$$V \xrightarrow{\wedge \eta} \bigwedge^{k+1} V,$$

we claim that $\eta = v_1 \wedge \cdots \wedge v_k$ for linearly independent $v_i \in V$ iff the kernel of this map has dimension at least k. If η is of this form then $v_1, \dots, v_k$ are in the kernel. Otherwise suppose $v_1, \dots, v_k$ are linearly independent vectors in the kernel. We may extend these v_i to a basis of V, so that there are scalars η_I such that

$$\eta = \sum_I \eta_I v_I, \qquad v_I = v_{i_1} \wedge \cdots \wedge v_{i_k} \text{ for } I = (i_1 < \cdots < i_k).$$

Then we know

$$0 = v_1 \wedge \cdots \wedge v_k \wedge \eta = \sum_I \eta_I v_1 \wedge \cdots \wedge v_k \wedge v_I = \sum_{I \cap [k] = \emptyset} \eta_I v_I,$$

so that for any multi-index I which is not $(1, \dots, k)$, we have $\eta_I = 0$. Thus η is a scalar multiple of $v_1 \wedge \cdots \wedge v_k$, as desired. And thus the image of the Grassmanian is

$$G(k, V) = \left\{ \eta \in \bigwedge^k V \;\middle|\; \operatorname{rank}(V \xrightarrow{\wedge \eta} \bigwedge^{k+1} V) \leq n - k \right\},$$

which is an algebraic subset of $\mathbb{P}(V^k)$ as it is the zeroset of all of the $(n-k+1)$st minors of the map $\bullet \wedge \eta$.

To show that $G(k, V)$ is irreducible, we note that there is a transitive action of $\mathrm{GL}(V)$ on $G(k, V)$. Choosing any $\Lambda \in G(k, V)$ we have that $\mathrm{GL}(V)\Lambda = G(k, V)$ is irreducible because it is the image of the irreducible $\mathrm{GL}(V)$ (it is an open subset of the irreducible $\mathbb{A}^n$). ■

Let $e_1, \dots, e_n$ be a basis for V and identify $V \cong \mathbb{F}^n$ so that Λ is the row space of some matrix $k \times n$ matrix $A = (a_{ij})$. Then $\bigwedge^k V$ is spanned by $e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$ for $I = (i_1 < \cdots < i_k)$. So for the basis $v_1, \dots, v_k \in \Lambda$ determined by A there are scalars p_I such that

$$v_1 \wedge \cdots \wedge v_k = \sum_I p_I e_I.$$

Then we see that p_I is the determinant of the submatrix of A whose columns are determined by I. These are the **Plücker coordinates** of A.

Example 1.3
Note that the Plücker embedding for $k = 1$ gives an isomorphism $G(1, V) \cong \mathbb{P}(V)$. So the most basic example of a Grassmanian is projective space.

Because $G(k, V)$ is a variety it has an affine cover. For an $(n-k)$-dimensional subspace $\Gamma \leq V$ define

$$U_\Gamma = \{\Lambda \in G \mid \Lambda \cap \Gamma = 0\}$$

to be the set of all subspaces which do not intersect Γ. This is a Zariski open subset of G, because taking $w_1, \ldots, w_{n-k}$ a basis for Γ and setting $\eta = w_1 \wedge \cdots \wedge w_{n-k}$, we see that

$$U_\Gamma = \left\{[\omega] \in G \subseteq \mathbb{P}(V^k) \;\middle|\; \omega \wedge \eta \neq 0\right\}.$$

We now claim that U_Γ is isomorphic to $\mathbb{A}^{k(n-k)}$. Let $\Omega \in U_\Gamma$ be a complementary k-space to Γ, so that $V = \Gamma \oplus \Omega$, and for any $\Lambda \in U_\Gamma$ we have projections

$$\Lambda \xrightarrow{\pi_\Gamma} \Gamma, \qquad \Lambda \xrightarrow{\pi_\Omega} \Omega$$

modulo Ω and Γ respectively. Because the kernel of π_Ω is Γ we see that π_Ω is an isomorphism, and so Λ is the graph of

$$\varphi\colon \Omega \xrightarrow{\pi_\Omega^{-1}} \Lambda \hookrightarrow V \xrightarrow{\pi_\Gamma} \Gamma.$$

Indeed its graph is

$$\operatorname{graph}\varphi = \left\{\omega + \pi_\Gamma \pi_\Omega^{-1}\omega \;\middle|\; \omega \in \Omega\right\},$$

and any $v \in \Omega$ is in this graph via $\omega = \pi_\Omega(v)$, and they have the same dimension.

Conversely if $\varphi\colon \Omega \to \Gamma$ is linear then its graph is complementary to Γ. So we have a correspondence

$$U_\Gamma \underset{\varphi \mapsto \operatorname{graph}\varphi}{\overset{\Lambda \mapsto \pi_\Gamma \circ \pi_\Omega^{-1}}{\rightleftarrows}} \hom(\Omega, \Gamma).$$

And considering them in Plücker coordinates, we see that these are morphisms of varieties. Because $\hom(\Omega, \Gamma) \cong \mathbb{A}^{k(n-k)}$, we see that $G(k, V)$ is a variety of dimension $k(n-k)$.

Proposition 1.4
The Grassmanian $G(k, V)$ is a projective variety of dimension $k(n-k)$.

Note:
In S. L. Kleiman and Dan Laksov, they define the Grassmanian $G(k, n)$ as the collection of k-planes in $\mathbb{P}^n$. This corresponds to our $G(k+1, n+1)$. We will denote this by

$$\mathbb{G}(k, V) = \mathbb{G}(k, \mathbb{P}(V)) \cong G(k+1, n+1).$$

2 The Chow Group

Definition 2.1
Consider a complex flag $\mathcal{V}$ in V, i.e. a sequence of subspaces

$$0 < V_1 < \cdots < V_{n-1} < V_n = V, \qquad \dim V_i = i.$$

Furthermore suppose we have a sequence a

$$n - k \geq a_1 \geq \cdots \geq a_k \geq 0.$$

Define the **Schubert cycle** to be the closed subset

$$\Sigma_a(\mathcal{V}) = \{\Lambda \in G(k,n) \mid \dim(V_{n-k+i-a_i} \cap \Lambda) \geq i \text{ for all } i\}.$$

Remark 2.2
In general for a subspace $\Gamma \leq V$, the set

$$\{\Lambda \in G(k,n) \mid \dim(\Gamma \cap \Lambda) \geq i\}$$

is closed, as $\dim(\Gamma \cap \Lambda) = \dim \Gamma + \dim \Lambda - \dim(\Gamma + \Lambda)$ so the condition is equivalent to $\dim(\Gamma + \Lambda) \leq \dim \Gamma + k - i$. Choosing bases for Λ, Γ, this is iff the rank of the matrix obtained by concatenating the bases has rank $\leq \dim \Gamma + k - i$, which is iff all submatrices of larger size have zero determinant.

For convenience, we write

$$\Sigma_{a_1,\ldots,a_s}(\mathcal{V}) = \Sigma_{a_1,\ldots,a_s,0,\ldots,0}$$

and

$$\Sigma_{a^s} = \Sigma_{a,\ldots,a}$$

to duplicate a s times.

Example 2.3
Note that for any flag $\mathcal{V}$,

$$\Sigma_0(\mathcal{V}) = G(k,n).$$

That is, any k-dimensional subspace Λ satisfies $\dim(V_{n-k+i} \cap \Lambda) \geq i$ for all i. Indeed,

$$\dim(V_{n-k+i} \cap \Lambda) = \dim V_{n-k+i} + \dim \Lambda - \dim(V_{n-k+i} + \Lambda) \geq n - k + i + k - n = i.$$

Example 2.4
Suppose V_ℓ is a subspace of dimension ℓ, then extend it to an arbitrary complete flag $\mathcal{V}$ and we see that

$$\Sigma_{n-k+1-\ell}(\mathcal{V}) = \{\Lambda \mid \Lambda \cap V_\ell \neq 0\}.$$

Indeed for $i = 1$ the Schubert condition gives

$$\dim(V_\ell \cap \Lambda) \geq 1 \iff V_\ell \cap \Lambda \neq 0.$$

The conditions on $i \geq 1$ are tautological as shown by the previous example.

Example 2.5

Given an ℓ-subspace V_ℓ then we have (for any extension $\mathcal{V}$)

$$\Sigma_{(n-\ell)^k}(\mathcal{V}) = \{\Lambda \mid \Lambda \subseteq V_\ell\},$$

because the Schubert conditions gives for all i

$$\dim(\Lambda \cap V_{\ell-k+i}) \geq i,$$

so for $i = k$ this means $\Lambda \subseteq V_\ell$, and then considering the first example the rest of the conditions follow.

And similarly

$$\Sigma_{(n-k)^\ell}(\mathcal{V}) = \{\Lambda \mid V_\ell \subseteq \Lambda\}$$

because

$$\dim(\Lambda \cap V_i) \geq i \text{ for } i \leq \ell \iff V_\ell \leq \Lambda$$

and the other conditions are tautological again.

Note that if we define for sequences $a \leq b$ iff $a_i \leq b_i$ for all i, then

$$\Sigma_a(\mathcal{V}) \subseteq \Sigma_b(\mathcal{V}) \iff a \leq b.$$

For a sequence a define

$$|a| = \sum_i a_i.$$

Theorem 2.6

The Schubert cycle $\Sigma_a(\mathcal{V})$ is an irreducible subvariety of $G(k, V)$ of codimension $|a|$.

Proposition 2.7

Let $e_1, \ldots, e_n$ be a basis for V and consider the standard flag $V_i = \langle e_1, \ldots, e_i \rangle$. Let a be a decreasing sequence and b be the strictly increasing sequence $b_i = n - k + i - a_i$. Then $\Sigma_a(\mathcal{V}) \subseteq G(k, n) \subseteq \mathbb{P}^N$ is the intersection

$$\Sigma_a(\mathcal{V}) = G(k, V) \cap \bigcap_{I \not\leq b} V(p_I),$$

where p_I is the Plücker coordinate associated to $I = (i_1 < \cdots < i_k)$.

Proof

For $\Lambda \in G(k,V)$ write Λ as the row space as the matrix $A = (a_{ij})$. The Plücker coordinate $p_I(\Lambda)$ is then the determinant of the submatrix of A whose columns are given by I.

Now suppose that $\Lambda \in \Sigma_a(V)$ so that $\dim(\Lambda \cap V_{b_i}) \geq i$ for all i. For $I \not\leq b$ there is a j such that $i_j > b_j$, and since $\dim(\Lambda \cap V_{b_j}) \geq j$, there are j linearly independent vectors in Λ whose coordinates past b_j are zero. If we take these vectors to be $v_1, \ldots, v_j$ and extend them to a basis $v_1, \ldots, v_k$ of Λ, then for every index in I past i_j (inclusive), the coordinates of $v_1, \ldots, v_j$ are zero. Thus the submatrix of A determined by I has a $j \times (k-j+1)$ block of zeroes. That is, it is of the form

$$\left(\begin{array}{c|c} * & 0 \\ \hline * & * \end{array}\right) \begin{matrix} j \\ k-j \end{matrix}.$$

Thus its rank is at most $(k-j)+(j-1) = k-1$ which means it is singular, so $p_I(\Lambda) = 0$.

Now suppose that $p_I(\Lambda) = 0$ for all $I \not\leq b$, but $\dim(\Lambda \cap V_{b_j}) < j$ for some j. So if consider the submatrix of A determined by columns of index $> b_j$, it must have rank $> k-j$. This is because there are $< j$ rows which are also in $\Lambda \cap V_{b_j}$:

$$\left(\begin{array}{c|c} * & 0 \\ \hline * & * \end{array}\right) \begin{matrix} < j \\ > k-j \end{matrix}$$

Performing row operations gives us a block-diagonal matrix of full rank, and so the right column of blocks is linearly independent. So there are at least $k-j+1$ columns with indices greater than b_j which are linearly independent. Extend these to k linearly independent columns (because A is of rank k) and let I be the indices of these columns. Then the submatrix determined by I is nonsingular so $p_I(\Lambda) \neq 0$. Because at least $k-j+1$ of these indices is greater than b_j, and as they are ascending this means that $i_j > b_j$, a contradiction. ■

The reason for wanting to consider Schubert cycles follows from the above examples. Consider the following problem.

Problem 2.8
If $V_1, V_2, V_3, V_4 \subseteq \mathbb{P}^{2n+1}$ are four general n-planes, how many lines $L \subseteq \mathbb{P}^{2n+1}$ intersect all four?

We know by the above example that $\Sigma_n(V_i) \subseteq \mathbb{G}(1, 2n+1)$ is the number of lines meeting V_i, so we want to count the intersection

$$Q = \bigcap \Sigma_n(V_i).$$

Unfortunately we do not yet have the tools to do this.

3 The Chow Groups

Definition 3.1

Let X be any algebraic variety. A **k-cycle** on X is a formal sum of subvarieties (closed integral subschemes) of X of dimension k. In other words the group of k-cycles on X is the free Abelian group on X's k-dimensional subvarieties, and is denoted $Z_k(X)$. X's **group of cycles** is the graded Abelian

group $Z(X) = \bigoplus_k Z_k(X)$.

A cycle $\sum n_i Y_i \in Z(X)$ is **effective** if all of the n_i are nonnegative. A **divisor** is an $(n-1)$-cycle (where $n = \dim X$).

For a variety X let $K(X)$ denote its field of rational functions. It is the fraction field of any one of its stalks, and is equal to its stalk at its generic point. For a subvariety $Y \subseteq X$, we denote by $\mathcal{O}_{X,Y}$ the stalk of X at Y's generic point.

Definition 3.2

Let $Z \subseteq X$ be a variety of codimension 1, then for $f \in K(X)^\times$ we denote its **order of vanishing** along Z as follows. Because $\mathcal{O}_{X,Z}$ is a one-dimensional local domain, for any nonzero $f \in \mathcal{O}_{X,Z}$ the quotient ring $\mathcal{O}_{X,Z}/(f)$ is a zero-dimensional Noetherian ring, and thus has finite length. We define f's order of vanishing to be

$$\operatorname{ord}_Z(f) = \operatorname{length}_{\mathcal{O}_{X,Z}} \mathcal{O}_{X,Z}/(f).$$

Any rational function $r \in K(X)^\times$ has the form f/g for $f, g \in \mathcal{O}_{X,Z}$ and so we define

$$\operatorname{ord}_Z(r) = \operatorname{ord}_Z(f) - \operatorname{ord}_Z(g).$$

It remains to show that the order of vanishing is well-defined. To do this we note that $\operatorname{ord}_Z(fg) = \operatorname{ord}_Z(f) + \operatorname{ord}_Z(g)$ for $f, g \in \mathcal{O}_{X,Z}$. This follows from the existence of the short exact sequence

$$0 \to \frac{\mathcal{O}_{X,Z}}{(f)} \xrightarrow{\times g} \frac{\mathcal{O}_{X,Z}}{(fg)} \xrightarrow{\text{proj}} \frac{\mathcal{O}_{X,Z}}{(g)} \to 0,$$

and the additivity of length. This also tells us that order is a homomorphism

$$\operatorname{ord}_Z \colon K(X)^\times \to \mathbb{Z}.$$

Note that if $f \notin \mathfrak{m}_{X,Z}$, meaning f does not vanish along Z, then f is invertible and so $\mathcal{O}_{X,Z}/(f) = 0$ and thus its order of vanishing is zero. This is of course an equivalence. Thus f has positive order of vanishing only when it vanishes along Z. And a rational function f/g may have a negative order of vanishing if its denominator vanishes ("more than its numerator") along Z.

Definition 3.3

Let $Y \subseteq X$ be a $(k+1)$-dimensional subvariety and let $r \in K(Y)^\times$. Define the k-cycle

$$\operatorname{div}(r) = \sum_Z \operatorname{ord}_Z(r) Z$$

where the sum runs over the codimension-1 subvarieties of Y. Let $\operatorname{Rat}_k(X)$ be the subgroup of $Z_k(X)$ generated by elements of the form $\sum_i \operatorname{div}(r_i)$ for $r_i \in K(Y_i)^\times$ where $Y_i \subseteq X$ is a $(k+1)$-dimensional subvariety.

Then define X's **Chow groups** by

$$\operatorname{Ch}_k(X) = \frac{Z_k(X)}{\operatorname{Rat}_k(X)}, \qquad \operatorname{Ch}(X) = \bigoplus_k \operatorname{Ch}_k(X).$$

Two cycles are **rationally equivalent** if they have the same image in $\operatorname{Ch}(X)$.